

**A COMPARATIVE ANALYSIS OF VOLTAGE OUTPUT - LEMON (CITRUS LIMON),
KALAMANSI (CITRUS MICROCARPA), POMELO (CITRUS MAXIMA), AND
ORANGE (CITRUS SINENSIS)**



Practical Research II

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CHAPTER I

Introduction

Background of the Study

Bioelectricity is an amazing study that is already done in the past. But as we continue to move along with the society, we also discovered a lot of studies with bioelectricity.

This generation of electrical power through biological processes, such as experimenting with how much electricity is produced by these citruses; Lemon (*Citrus limon*), Kalamansi (*Citrus microcarpa*), Pomelo (*Citrus maxima*), and Orange (*Citrus sinensis*). Citrus fruits are found all over the place, each with its own unique traits that might affect how they produce bioelectricity. In this study, we're diving into a detailed comparison of the voltage output from Lemon, Kalamansi, Pomelo, and Orange, aiming to uncover any interesting differences in how they generate bioelectricity.

The objective of the study is to create a source of energy by using citrus fruits. Our team aims to enhance the study about the properties of citrus fruits and make it better for further study.

In 1791, Galvani initially postulated that electricity was exclusive to animal origin and living tissues. Subsequently, in 1800, Volta extended the understanding by

demonstrating that organic substances could generate electricity. Volta's groundbreaking experiment involved a setup comprising a small sheet of copper, zinc, and cloth spacers soaked in an acid solution, marking the inception of electricity production (Gollner & Adam, 2010). The readily available and relatively safe zinc and copper electrodes were employed, showcasing the varied voltages produced by copper, zinc, and other metals such as lead, iron, and magnesium (Abdullah, 2018).

Wani, M. (2018) contributed insights into electrolytes, emphasizing their derivation from acids dissolved in water, as effective electricity conductors when incorporated into a circuit. Notably, electricity fails to conduct in waterless states, including dry, powdered, or crystalline forms, even when exhibiting the same acidity level. As a concluding observation, Wani highlighted that acids act as electron producers, while water serves as the conductor.

Deziel, C. (2018) reported on the abundance of liquid-form juice in citrus fruits, particularly oranges or lemons, which demonstrated acidity when tested with zinc and copper strips. The experiment illustrated the acidic nature of the juice, containing electrolytes capable of generating electricity. This phenomenon allows two different metals to react, as the fruit alone cannot produce electricity. Furthermore, it was asserted that local fruits with high acidic content could harness this method for electricity generation, extending beyond citrus fruits.

The purpose of this research is to determine which citrus and local fruits would produce the most electricity flow. The citrus and local fruits should contain high acidity content to conduct electricity and capable to be one of the power sources.

Statement of the Problem

Our investigation delves into a comparative analysis of voltage output from Lemon (Citrus limon), Kalamansi (Citrus microcarpa), Pomelo (Citrus maxima), and Orange (Citrus sinensis) fruits, exploring their potential as sources of electrical energy. We aim to address the following research questions:

1. How does the amount of electrical energy differ when we use lemons, kalamansi, pomelos, and oranges?
2. What things like the size, ripeness, and sourness of these fruits affect how much electrical energy they can produce?
3. What can we use the differences in electrical energy from lemons, kalamansi, pomelos, and oranges for, especially in making new energy sources and technology better?

These research questions will guide the investigation into the comparative analysis of voltage output among different citrus fruits and contribute to understanding their potential in all other aspects such as; alternative electricity and help for the further studies.

Definition of Terms

Abundance - refers to a plentiful or ample quantity or supply of something.

Bioelectricity - is the electrical energy produced by living organisms, including the electricity generated from organic materials like citrus fruits for various purposes, such as power generation.

Circuit - refers to a closed loop or pathway through which electricity can flow, typically consisting of conductive materials and components such as wires, resistors, capacitors, and power sources.

Citrus- refers to a genus of flowering plants in the rue family, Rutaceae, which includes fruits such as oranges, lemons, limes, and grapefruits.

Electrolytes - are substances that conduct electricity when dissolved in water or melted, essential for various physiological processes in living organisms.

Groundbreaking - refers to something innovative or pioneering, often describing a significant advancement or achievement that sets a new standard or breaks new ground in a particular field or context.

Harness - means to utilize or control something, typically for a specific purpose or task.

Postulated - means to suggest or hypothesize something based on available evidence or reasoning.

This is partial paper